CUSTOMER SEGMENTATION AND CLUSTERING ANALYSIS

Mall Customer Dataset | Exploratory Data Analysis & K-Means Clustering

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Tools Used: Python | Pandas | Seaborn | Scikit-Learn | Matplotlib

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Executive Summary

This report presents a comprehensive customer segmentation analysis performed on the Mall Customers dataset, containing 200 customer records with attributes including Gender, Age, Annual Income, and Spending Score. The analysis follows a structured data science workflow, from data loading and cleaning through exploratory data analysis (EDA) to K-Means machine learning clustering — to identify distinct customer segments that can drive targeted marketing strategies.

Key Findings

- The dataset contains 200 customers (112 Female, 88 Male) with zero missing or duplicate values.
- Customer ages range from 18 to 70 years, with a median age of 36 years.
- Annual incomes span $15,000 to $137,000, with an average of approximately $60,560.
- Spending scores range from 1 to 99, with a near-perfectly balanced average of 50.2.
- There is virtually no linear correlation (r = 0.01) between annual income and spending score, confirming that income alone does not predict spending behaviour.
- Age shows a moderate negative correlation (r = -0.33) with spending score, suggesting younger customers tend to spend more.
- K-Means clustering (k = 5, optimal by Elbow Method) identified four actionable business segments.

Business Segments Identified

Segment	Avg Age	Avg Income (k$)	Avg Spend Score	Size
High Income – High Spender	32.88	86.10	81.53	40 (20.0%)
High Income – Low Spender	44.39	89.77	18.48	31 (15.5%)
Low Income – High Spender	25.77	26.12	74.85	26 (13.0%)
Low Income – Low Spender	42.81	50.54	41.36	103 (51.5%)

1. Project Introduction

1.1 Project Overview

Customer segmentation is the process of grouping customers into subsets based on shared characteristics such as demographics, income, and purchasing behaviour. This project applies unsupervised machine learning, specifically K-Means clustering, to the Mall Customers dataset to identify natural groupings of shoppers and extract actionable business intelligence.

1.2 Objectives

1. Load, inspect, and clean the Mall Customers dataset.
2. Conduct univariate and bivariate exploratory data analysis (EDA).
3. Apply K-Means clustering to identify customer segments.
4. Profile and label each cluster with business-friendly segment names.
5. Derive marketing and strategic recommendations from the findings.

1.3 Tools and Libraries

Library / Tool	Purpose
pandas 2.3.3	Data loading, cleaning, and manipulation
numpy 2.3.5	Numerical operations and array handling
matplotlib	Base data visualisation library
seaborn 0.13.2	Statistical charts and visualisations
scipy.stats	Pearson correlation and statistical testing
sklearn.cluster.KMeans	K-Means clustering algorithm
sklearn.preprocessing.StandardScaler	Feature standardisation / normalisation
sklearn.decomposition.PCA	Dimensionality reduction (advanced analysis)
sklearn.metrics.silhouette_score	Cluster quality evaluation

1.4 Dataset Overview

The dataset used is Mall_Customers.csv, a widely recognised academic dataset for practising customer segmentation. It contains 200 rows and 5 columns with no missing values or duplicate records.

Column	Data Type	Range / Values	Description
CustomerID	int64	1 – 200	Unique customer identifier
Gender	category	Male, Female	Gender of the customer
Age	int64	18 – 70 years	Age of the customer
Annual Income (k$)	int64	$15k – $137k	Annual income in thousands USD
Spending Score (1-100)	int64	1 – 99	Mall-assigned spending behaviour score

2. Data Cleaning and Preprocessing

2.1 Data Quality Assessment

Before any analysis, a thorough data quality check was conducted. The following checks were performed and passed without requiring remediation:

Quality Check	Result	Action Taken
Missing Values	0 across all 5 columns	No action required
Duplicate Records	0 duplicates detected	No action required
Data Types	4 int64, 1 object (Gender)	Gender converted to 'category' type
Outliers – Age	No outliers (IQR range: 28.75 – 49.0)	No action required
Outliers – Annual Income	1 mild outlier at $137k	Retained – valid data point
Outliers – Spending Score	No outliers detected	No action required

2.2 Column Renaming

All column headers were standardised to lowercase snake_case format to improve code readability and reduce the risk of reference errors during analysis:

Original Name	Renamed To
CustomerID	**customer_id**
Gender	**gender**

Age	**age**
Annual Income (k$)	**annual_income**
Spending Score (1-100)	**spending_score**

2.3 Data Type Optimisation

The gender column was converted from a generic object type to the pandas category dtype. This optimisation reduces memory usage, improves grouping performance, and makes the column semantically correct for downstream categorical analysis.

2.4 Missing Value Treatment (Reference Framework)

Although this dataset contained no missing values, the following standard treatment methods were documented for reference:

- df.dropna() — Remove rows with any missing values.
- df['age'].fillna(df['age'].mean(), inplace=True) — Replace missing numeric values with the column mean.
- df['annual_income'].fillna(df['annual_income'].median(), inplace=True) — Replace with median (preferred for skewed distributions).

3. Exploratory Data Analysis (EDA)

EDA was conducted in two phases: Univariate Analysis (examining each variable individually) and Bivariate Analysis (examining relationships between pairs of variables).

3.1 Univariate Analysis

3.1.1 Gender Distribution

Gender	Count	Percentage
Female	112	56.0%
Male	88	44.0%
Total	**200**	**100%**

Insight: Female customers constitute the majority of mall visitors at 56%, compared to 44% male. This gender skew suggests that marketing strategies and product offerings targeted towards female shoppers are likely to have a greater impact on overall mall revenue.

3.1.2 Age Distribution

Statistic	Value	Interpretation
Mean Age	38.85 years	Average customer is ~39 years old
Median Age	36.00 years	50% of customers are below 36
Std Deviation	13.97 years	Moderate spread around the mean
Minimum Age	18 years	Youngest customer in dataset
Maximum Age	70 years	Oldest customer in dataset
IQR (25th – 75th)	28.75 – 49.00 years	Core customer age band

Insight: The age distribution is roughly right-skewed, with the highest concentration of customers in the 30–35 age range. The boxplot confirms no significant outliers. The mall attracts a broad demographic spanning 18 to 70 years, with the core customer base aged 29 to 49.

3.1.3 Annual Income Distribution

Statistic	Value (k$)	Interpretation
Mean Income	$60,560	Average customer earns ~$60k/year
Median Income	$61,500	Near-symmetric distribution
Std Deviation	$26,265	Wide spread across income levels
Minimum Income	$15,000	Lowest earner in dataset
Maximum Income	$137,000	Highest earner (mild outlier)
IQR (25th – 75th)	$41,500 – $78,000	Middle 50% income band

Insight: The income distribution is approximately bell-shaped and centred around $60,000. The mall primarily attracts middle-income customers, although both lower-income and higher-income shoppers are present. One mild outlier at $137,000 was retained as a valid data point.

3.1.4 Spending Score Distribution

Statistic	Value	Interpretation
Mean Score	50.20	Perfectly centred spending distribution
Median Score	50.00	Symmetrical distribution confirmed
Std Deviation	25.82	High variance across customer types
Minimum Score	1	Extremely low spenders present
Maximum Score	99	Near-maximum high spenders present
IQR (25th – 75th)	34.75 – 73.00	Moderate to moderately-high spenders

Insight: The spending score is uniformly distributed between 1 and 99, with an average of 50.2. The boxplot shows no outliers, indicating that all spending patterns fall within expected normal ranges. This even distribution suggests the dataset captures a well-balanced mix of budget-conscious and free-spending customers.

3.2 Bivariate Analysis

3.2.1 Annual Income vs. Spending Score

The scatter plot of annual income versus spending score is the most critical visualisation in this project. It reveals that income level alone does not determine spending behaviour. The Pearson correlation coefficient between the two variables is $r = 0.01$ (p-value = 0.889), indicating virtually no linear relationship.

However, the scatter plot reveals five distinct natural clusters of customers, later confirmed by K-Means clustering:

- Top-right cluster: High income & high spending — the premium customer segment.
- Bottom-right cluster: High income & low spending — conservative high earners.
- Top-left cluster: Low income & high spending — impulsive or brand-loyal shoppers.
- Bottom-left cluster: Low income & low spending — budget-conscious customers.
- Central cluster: Middle income & average spending — the regular mainstream shoppers.

3.2.2 Age vs. Spending Score

The correlation analysis between age and spending score yields r = -0.33, indicating a moderate negative relationship. As customer age increases, spending score tends to decrease. The regression trendline confirms this downward pattern.

Age Group	Spending Behaviour Observation
18–30 years	Highest concentration of high spenders (score 70–99); impulsive & trend-driven purchasing.
31–45 years	Mixed spending scores (30–70); transitioning from impulsive to moderate behaviour.
46–70 years	Predominantly moderate to low spenders (10–50); more conservative and planned purchasing.

4. Customer Segmentation – K-Means Clustering

4.1 Preprocessing for Clustering

Three features were selected for clustering: age, annual_income, and spending_score. Since these features are on different scales (years, thousands of dollars, and a 1–100 index), StandardScaler was applied to normalise each feature to zero mean and unit variance before fitting the clustering model. This prevents any single feature from dominating the distance calculations.

4.2 Optimal Number of Clusters – Elbow Method

The Elbow Method was applied by fitting K-Means models for k = 1 through k = 10 and plotting the within-cluster sum of squares (inertia) against the number of clusters. The inertia decreases sharply from k = 1 to k = 4 and begins to flatten after k = 5, forming a clear 'elbow'. Based on this, k = 5 was selected as the optimal number of clusters, balancing model complexity with cluster separation quality.

k Value	Inertia (approx.)	Observation
1	600	All customers in one group – no segmentation
2	390	Large drop – first meaningful split
3	295	Continued improvement
4	207	Significant drop continues

5	**168**	**Elbow point – optimal balance (SELECTED)**
6	156	Diminishing returns begin
7–10	<135	Marginal gains – over-segmentation risk

4.3 Cluster Profiling Results

After fitting K-Means with k = 5 on the standardised features, each customer was assigned a cluster label. The cluster centroids were then back-mapped to the original feature space to compute average profile metrics per segment:

Cluster	Business Label	Avg Age	Avg Income (k$)	Avg Spend Score	Size
0	**Low Income – Low Spender (Mainstream)**	55.28	47.62	41.71	58 (29%)
1	**High Income – High Spender (Premium)**	32.88	86.10	81.53	40 (20%)
2	**Low Income – High Spender (Impulsive)**	25.77	26.12	74.85	26 (13%)
3	**Low Income – Moderate Spender (Regular)**	26.73	54.31	40.91	45 (22.5%)
4	**High Income – Low Spender (Conservative)**	44.39	89.77	18.48	31 (15.5%)

5. Detailed Segment Profiles

Segment 1 – High Income, High Spender (Premium Customers)

Attribute	Detail
Average Age	32.88 years
Average Annual Income	$86,100
Average Spending Score	81.53 / 100
Cluster Size	40 customers (20% of total)

Customer Profile	Young, high-earning professionals who actively spend at the mall. They are the most valuable customer segment.
Marketing Strategy	Offer premium loyalty programmes, exclusive early access to products, VIP events, and personalised shopping experiences.

Segment 2 – High Income, Low Spender (Conservative Customers)

Attribute	Detail
Average Age	44.39 years
Average Annual Income	$89,770
Average Spending Score	18.48 / 100
Cluster Size	31 customers (15.5% of total)
Customer Profile	Middle-aged high earners who visit the mall infrequently or make conservative purchasing decisions. High untapped potential.
Marketing Strategy	Deploy targeted retention campaigns, exclusive high-value promotions, and personalised product recommendations to convert conservative behaviour.

Segment 3 – Low Income, High Spender (Impulsive Customers)

Attribute	Detail
Average Age	25.77 years
Average Annual Income	$26,120
Average Spending Score	74.85 / 100
Cluster Size	26 customers (13% of total)
Customer Profile	Young, lower-income customers who spend heavily relative to their earnings. Likely driven by trends, social influence, and lifestyle aspirations.
Marketing Strategy	Offer instalment payment plans, flash sales, social-media-driven promotions, and value-for-money bundles to sustain engagement.

Segment 4 – Low Income, Low Spender (Mainstream Customers)

Attribute	Detail
Average Age	42.81 years
Average Annual Income	$50,540
Average Spending Score	41.36 / 100
Cluster Size	103 customers (51.5% of total)
Customer Profile	The largest segment. Middle-aged, mid-to-lower income customers with average spending behaviour. Represent the mall's primary foot traffic.
Marketing Strategy	Budget-friendly promotions, loyalty discount schemes, seasonal sales events, and BOGO (Buy One Get One) offers to drive volume.

6. Business Recommendations

6.1 Marketing Campaign Strategies

Based on the four identified customer segments, the following targeted marketing approaches are recommended:

Segment	Recommended Campaign	Expected Outcome
High Income – High Spender	VIP loyalty rewards, premium product launches, exclusive events	Increase visit frequency and average basket size
High Income – Low Spender	Personalised offers, targeted re-engagement emails, experience-based incentives	Convert high-earning but passive visitors into active spenders
Low Income – High Spender	Instalment plans, BNPL options, trend-based flash deals	Sustain high-frequency spending while protecting financial wellbeing
Low Income – Low Spender	Discount codes, bundle deals, seasonal promotions, value-range spotlights	Increase spend per visit through perceived value

6.2 Strategic Planning

- Allocate marketing budget proportionally to segment size and revenue potential. The High Income – High Spender segment (20%) should receive disproportionately high investment relative to size, given its high revenue per customer.
- Focus retention efforts on the High Income – Low Spender group — this is the highest-value untapped segment. Re-engaging even 20% of this group could yield significant revenue uplift.
- Leverage social media and influencer-based campaigns to engage the Low Income – High Spender group, who are young and trend-sensitive.
- Design mall-wide seasonal promotions (e.g., Black Friday, holiday sales) primarily targeting the Mainstream segment, which comprises over 51% of the customer base.

6.3 Sales Forecasting Implications

Segment-based forecasting enables more accurate revenue projections. By tracking changes in segment membership over time (e.g., shifts in average spending score or income bracket), the mall can anticipate demand fluctuations and plan inventory and staffing accordingly.

7. Conclusions

7.1 Summary of Findings

This project successfully completed a full end-to-end customer segmentation pipeline on the Mall Customers dataset. Starting from raw data, the workflow covered data loading, quality verification, cleaning, exploratory analysis, and unsupervised machine learning clustering to produce four actionable business segments.

The key analytical outcomes are:

1. Annual income and spending score have virtually zero linear correlation ($r = 0.01$), meaning higher earners are not necessarily bigger spenders.
2. Age shows a moderate negative correlation with spending score ($r = -0.33$), indicating younger customers are more active spenders.
3. Female customers represent 56% of mall shoppers, suggesting a female-skewed primary audience.
4. K-Means with $k = 5$ (selected via Elbow Method) produced well-separated, interpretable clusters.
5. The largest segment — Low Income, Low Spender — comprises 51.5% of customers and represents the mall's primary foot-traffic base.
6. The highest-value untapped opportunity lies in the High Income, Low Spender segment, which earns the most ($89,770 average) but spends the least.

7.2 Limitations

- The dataset contains only 200 records. While sufficient for academic demonstration, a real-world deployment would benefit from larger, longitudinal datasets.
- The analysis does not account for temporal patterns (e.g., seasonal spending shifts) or external factors (economic conditions, promotions history).
- The Spending Score is a mall-assigned metric; its exact calculation methodology is not disclosed, which limits interpretability.

7.3 Future Work

- Apply DBSCAN or Hierarchical Clustering to compare segment boundaries against K-Means results.
- Incorporate additional features (e.g., visit frequency, product category preferences) to enrich segment profiles.
- Deploy a real-time customer segmentation dashboard using Streamlit or Power BI to allow dynamic segment tracking.
- Explore predictive modelling to forecast which segment a new customer belongs to upon first visit.

8. Appendix – Statistical Reference

A. Descriptive Statistics Summary

Statistic	Age	Annual Income (k$)	Spending Score	Customer ID
Count	200	200	200	200
Mean	38.85	60.56	50.20	100.5
Std Dev	13.97	26.26	25.82	57.88
Min	18	15	1	1
25th Percentile	28.75	41.50	34.75	50.75
Median (50th)	36.00	61.50	50.00	100.5
75th Percentile	49.00	78.00	73.00	150.25
Max	70	137	99	200

B. Correlation Summary

Variable Pair	Pearson r	p-value	Interpretation
Annual Income vs Spending Score	0.01	0.889	No linear relationship
Age vs Spending Score	-0.33	< 0.05	Moderate negative relationship

End of Report | Customer Segmentation and Clustering Analysis | Portfolio Project